

WEIGHTS IN THE COHOMOLOGY OF ALGEBRAIC VARIETIES

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1. Let X be a complex algebraic variety, that is, a separated scheme of finite type over $\mathbf{C}$. We again write X for the usual topological space $X(\mathbf{C})$ underlying it. In this lecture I describe a remarkable filtration of the rational cohomology groups of X , the *weight filtration*, and give a collection of results about its properties. Its definition will be given in §12. For proofs, I refer to the works cited in the bibliography, where the theorems are often proved in more general settings. Working over $\mathbf{Q}$ allows one to have at the same time Hodge theory, actions of Galois groups, and resolution of singularities.

The weight filtration is a finite increasing filtration. I denote it by W . It is also defined in relative situations and in cohomology with proper supports. It depends not only on the topological space X , but also on the way in which that space is realized as an algebraic variety. It is compatible with Künneth isomorphisms: under the isomorphism

$$H^*(X \times Y) = H^*(X) \otimes H^*(Y)$$

one has

$$W_j(H^*(X \times Y)) = \sum_{k+l=j} W_k(H^*(X)) \otimes W_l(H^*(Y)).$$

It is functorial for algebraic morphisms. More precisely, if $f : X \rightarrow Y$ is algebraic, then

$$f^* : H^*(Y) \rightarrow H^*(X)$$

is strictly compatible with the weight filtrations of $H^*(Y)$ and $H^*(X)$.¹ More generally, every natural map is strictly compatible with the weight filtrations.

The weight filtration is a discrete invariant: it is invariant under deformation of the algebraic structure of X . More precisely, one has the following theorem.

THEOREM 1. *Let $f : X \rightarrow S$ be a morphism. For $t \in S$, let $X_t = f^{-1}(t)$. If the sheaf $R^i f_* \mathbf{Q}$ is locally constant, that is, a local system, then there is a weight filtration W of $R^i f_* \mathbf{Q}$ by local subsystems such that the maps*

$$r_t : (R^i f_* \mathbf{Q})_t \rightarrow H^i(X_t)$$

are strictly compatible with the weight filtrations, and in particular such that r_t is an isomorphism and W induces the weight filtration of $H^i(X_t)$.

3. Roughly speaking, the weight filtration expresses how the cohomology of X may be built from the cohomology of nonsingular projective varieties. Here are several examples.

EXAMPLE 2 (3.1). If X is proper, equivalently compact for example projective, and smooth, equivalently nonsingular, then $H^i(X)$ is pure of weight i :

$$\mathrm{Gr}_j^W H^i(X) = 0 \quad (j \neq i).$$

Equivalently, $W_{i-1} H^i(X) = 0$ and $W_i H^i(X) = H^i(X)$.

EXAMPLE 3 (3.2). Let X be smooth, and let X be open in a nonsingular compactification $\overline{X}$ with complement a divisor with normal crossings. If the irreducible components of $\overline{X} - X$ are denoted Y_α , and if $Y_I = \bigcap_{\alpha \in I} Y_\alpha$, then the graded pieces of $H^n(X)$ are controlled by the cohomology of the Y_I . In particular the possible weights of $H^n(X)$ lie between n and $2n$.

¹If a class $x \in H^*(X)$ lies in the image of f^* , then it has filtration $\leq i$ if and only if it is the image of a class of filtration $\leq i$.

EXAMPLE 4 (3.3). If X is proper, possibly singular, then the possible weights of $H^n(X)$ are at most n .

EXAMPLE 5 (3.4). If X is arbitrary, then the possible weights of $H^n(X)$ lie between 0 and $2n$.

4. Let H be a finite-dimensional vector space over $\mathbf{Q}$, provided with a finite increasing filtration W . A *grading* of W is a direct-sum decomposition

$$H = \bigoplus H_i$$

such that

$$W_j H = \bigoplus_{i \leq j} H_i.$$

The filtration W is always gradable, but noncanonically. In the situations considered above, it is often useful to know more: the grading can sometimes be chosen to respect additional algebraic structures.

5. Suppose, for example, that H is the cohomology of a differential graded algebra $\mathcal{M}$, or of a minimal model in rational homotopy theory. One seeks a grading compatible not only with the vector spaces H^i , but also with products and differentials. In favorable cases, one obtains a grading of the whole model whose induced grading on cohomology is the weight grading. Such a grading imposes severe restrictions on the rational homotopy type.

6. Let X be smooth. Then $H^n(X)$ is controlled by the part of the square $[0, n] \times [0, n]$ above the second diagonal, in the following sense. The mixed Hodge structure on $H^n(X)$ has nonzero Hodge numbers only for pairs (p, q) in a region dictated by smoothness. In particular the weights range from n to $2n$.

7. More explicitly, for the cohomology groups considered below, the following regions $\mathcal{C}$ in the square $[0, n] \times [0, n]$ control the possible Hodge numbers and weights.

(7.1) If X is proper, $H^n(X)$ is controlled by

$$\{(p, q) \in [0, n] \times [0, n] \mid p + q \leq n\}.$$

(7.2) If X is smooth, $H^n(X)$ is controlled by

$$\{(p, q) \in [0, n] \times [0, n] \mid p + q \geq n\}.$$

(7.3) If $N = \dim X \leq n$, the square $[0, n] \times [0, n]$ may be replaced by the square $[n - N, N] \times [n - N, N]$.

8. Variants.

(8.1) The group with proper supports $H_c^n(X)$ is controlled by the part of the square $[0, n] \times [0, n]$ below the second diagonal. For $N = \dim X \leq n$, this square may again be replaced by $[n - N, N] \times [n - N, N]$. In particular $H_c^{2N}(X)$ is pure of weight $2N$.

(8.2) The “duality” between the proper and smooth cases follows from Poincaré duality. If X is smooth and purely of dimension N , then

$$H^n(X) = \text{Hom}(H_c^{2N-n}(X), H_c^{2N}(X)) = (H_c^{2N-n}(X)^\vee)(-N).$$

This argument extends the results stated for smooth X to the case where X is a rational homology manifold. For such an X , the image of $H_c^n(X)$ in $H^n(X)$ is pure of weight n .

(8.3) If X is assumed only normal, it remains true that $H^1(X)$ is controlled by

$$\{(0, 1), (1, 0), (1, 1)\}.$$

9. Let X be a connected variety, let $x \in X$, and let $\Gamma^{(n)}$ be the largest torsion-free nilpotent quotient of length n of $\pi_1(X, x)$. It is known that sufficiently small finite-index subgroups H of $\Gamma^{(n)}$ have the following property.

(*) There exists a nilpotent Lie algebra of length n , $(\mathcal{G}, [])$, with $\mathcal{G}$ a free $\mathbf{Z}$ -module, such that the Campbell–Hausdorff formula

$$x \circ y = x + y + [x, y]/2 + [x, [x, y]]/12 + [y, [y, x]]/12 + \cdots$$

turns $\mathcal{G}$ into a group isomorphic to H .

Moreover, the Lie algebra

$$\mathcal{G}^{(n)} = \mathcal{G} \otimes \mathbf{Q}$$

depends only on $\Gamma^{(n)}$, not on H .

THEOREM 6 (Morgan). *If X is normal, then $\mathcal{G}^{(n)}$ admits a grading W , in negative degrees, such that $\mathcal{G}^{(n)}$ is generated by elements of degrees -1 and -2 , subject only to relations of degrees -2 , -3 , and -4 , together with the vanishing of n -fold iterated commutators.*

It is more convenient to work with the projective system $\mathcal{G}^{(\infty)}$ of the $\mathcal{G}^{(n)}$. Defining

$$H^i(\mathcal{G}^{(\infty)}) = \varinjlim H^i(\mathcal{G}^{(n)}),$$

one has

$$H^1(\mathcal{G}^{(\infty)}) = H^1(X), \quad H^2(\mathcal{G}^{(\infty)}) \subset H^2(X).$$

Furthermore, a grading E of $\mathcal{M}$ as in §5 defines compatible gradings of $\mathcal{G}^{(\infty)}$, of its cohomology, and of the cohomology of X . By (8.3), respectively (7.1), $H^1(X)$ has weights 1 and 2, and $H^2(X)$ has weights at most 4. Hence $H^1(\mathcal{G}^{(\infty)})$, the dual of the group $\mathcal{G}^{(\infty)}/[\mathcal{G}^{(\infty)}, \mathcal{G}^{(\infty)}]$ of generators of $\mathcal{G}^{(\infty)}$, has weights 1 and 2, while $H^2(\mathcal{G}^{(\infty)})$, the dual of the group of relations among the generators, has weights at most 4. This proves the theorem.

11. For a proper smooth variety, the weight filtration reduces to very little (3.1). The fact that it can be graded in the sense of §4 implies the vanishing of all Massey products. The fact that it can be graded in the sense of §5 even implies that the rational homotopy type of X can be read off from its cohomology algebra; see [?].

12. Hodge theory. The cohomology group $H^i(X)$ of every complex algebraic variety is endowed with a mixed Hodge structure (W, F) . Here W is a filtration of $H^i(X)$ and F is a filtration of

$$H^i(X) \otimes \mathbf{C} = H^i(X, \mathbf{C}),$$

with W and F satisfying suitable axioms; see [?]. The weight filtration is, by definition, this filtration W .

13. l -adic theory. An algebraic variety X is defined by finitely many polynomial equations. The coefficients of these equations generate a subring R of $\mathbf{Q}$ of finite type, with fraction field K , and X is obtained by extension of scalars to $\mathbf{C}$ from a scheme over K , indeed over R . The statements below become true after replacing R by $R[1/f]$, with $f \in R$ sufficiently divisible.

- (13.1) Let l be a prime number. The group $H^*(X) \otimes \mathbf{Q}_l$ identifies with the l -adic cohomology group $H^*(X, \mathbf{Q}_l)$. The latter is defined in a purely algebraic way, and therefore, by transport of structure, carries an action of $\text{Aut}(\mathbf{C}/K)$.
- (13.2) Let $\overline{K}$ be the algebraic closure of K inside $\mathbf{C}$. The action factors through $\text{Gal}(\overline{K}/K)$. It is unramified over $R[1/l]$, that is, it factors through the Galois group of the largest extension $K_{\text{nr}} \subset \overline{K}$ of K unramified over $R[1/l]$.
- (13.3) Let $\mathfrak{m}$ be a maximal ideal of $R[1/l]$, and put $N(\mathfrak{m}) = \#R/\mathfrak{m}$, so that $R/\mathfrak{m}$ is a finite field. To $\mathfrak{m}$ corresponds a conjugacy class of Frobenius substitutions $\varphi_{\mathfrak{m}} \in \text{Gal}(K_{\text{nr}}/K)$, whose inverses are the geometric Frobenius elements

$$F_{\mathfrak{m}} = \varphi_{\mathfrak{m}}^{-1}.$$

THEOREM 7. *For f sufficiently divisible, the following hold.*

- (i) *For every l and every $\mathfrak{m}$, the eigenvalues α of $F_{\mathfrak{m}}$, considered in a suitable finite extension of $\mathbf{Q}_l$, are algebraic integers. For each α , there is an integer $w(\alpha)$ such that all complex conjugates of α have absolute value*

$$N(\mathfrak{m})^{w(\alpha)/2}.$$

- (ii) *Choose a geometric Frobenius $F_{\mathfrak{m}}$, and let ${}_{\mathfrak{m}}W_j$ be the sum of the generalized eigenspaces corresponding to eigenvalues α of $F_{\mathfrak{m}}$ such that $w(\alpha) = j$. The filtration by the ${}_{\mathfrak{m}}W_j$ is independent of $\mathfrak{m}$ and of the choice of $F_{\mathfrak{m}}$; it is rational and its trace on $H^*(X)$ is independent of l . It is the weight filtration.*

PRINCIPLE OF THE PROOF. One expresses the cohomology of X , by means of a spectral sequence, in terms of the cohomology of proper smooth varieties, as in [?]. The spectral sequence converges to a filtration W of $H^*(X)$, which is by definition the weight filtration in Hodge theory. This spectral sequence has an l -adic analogue, so the l -adic extension W_l of W is stable under a Galois group. The Weil conjecture implies that the eigenvalues of $F_{\mathfrak{m}}$ on $\text{Gr}_j^{W_l}$ are algebraic integers with complex absolute values $N(\mathfrak{m})^{j/2}$, and the theorem follows.

15. In the cases considered in (7.1), (7.2), and (7.3), the region $\mathcal{C}$ described there controls not only the weights but also the Hodge numbers and the divisibility of the eigenvalues of the $F_{\mathfrak{m}}$. More precisely, in (7.1), (7.2), and (7.3), a region $\mathcal{C}$ was assigned to a group H , and:

- (a) the nonzero Hodge numbers $h^{p,q}$ of the Hodge structures $\mathrm{Gr}_j^W(H)$ satisfy $(p, q) \in \mathcal{C}$;
- (b) useful only in case (7.3), if the pairs $(p, q) \in \mathcal{C}$ with $p + q = j$ satisfy $p, q \geq k$, then, again for f sufficiently divisible, the eigenvalues of $F_{\mathfrak{m}}$ with complex absolute value $N(\mathfrak{m})^{j/2}$ are divisible by $N(\mathfrak{m})^k$.

One hopes that this is a special case of a general principle; see [?].

16. The following result is clear from the viewpoint of Hodge theory.

(16.1) For j odd, $\dim \mathrm{Gr}_j^W H^n(X)$ is even.

By contrast, I do not yet know how to prove §§2 and 5 except by the l -adic method. For §5, the method of §14 gives an l -adic grading; a trick of Sullivan then implies the existence of rational gradings of the required type.

The functoriality properties of the weight filtration are evident from the l -adic point of view, because Galois commutes with everything that can be defined. It is nevertheless useful to prove them from the Hodge-theoretic point of view, in order to obtain analogous properties for the filtration F . Morgan has obtained many results in this direction, enough to prove a slightly weaker form of §10 by Hodge theory.

17. As §5 suggests, a weight filtration also exists on the homotopy groups, after tensoring with $\mathbf{Q}$, under hypotheses of simple connectedness.

It also exists on groups of vanishing cycles (cf. [?]), and I hope that these filtrations will help us understand them better.

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